Local Validity and Contextual Holonomy: A Structural Formulation of the Copenhagen Principle

(v3 — complete rewrite)

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Abstract

We propose that the Copenhagen viewpoint, stripped of its rhetoric, is a set of structural statements about one geometric object: the transformation groupoid $\mathrm{PU}(n) \ltimes X$ of measurement contexts (maximal abelian subalgebras) with its canonical U(1) 2-cocycle. In this formulation: *local validity* — the exact classicality of every single context — is a theorem with an explicit witness; *complementarity* is the nonvanishing holonomy of the cocycle along closed context loops, which detects exactly the all-versus-nothing sector of contextuality (an equivalence, now proved), with the achievable state-independent values classified (the value classification — even/odd dichotomy and fixed value $d/2$ — is due to Abramsky, Cercelescu & Constantin, FSCD 2024, and is recast here in certificate language) and the state sector governed by a facet geometry that is now completely enumerated in the certified $d = 4$ example (23,256 facets in 61 symmetry classes) and classified in closed form at $d = 2$; *collapse* is pinned by statistics, repeatability, and covariance to a one-parameter family per degenerate block whose unique efficient member is the Lüders rule — with a classical twin theorem identifying the same axioms' classical solution as Bayesian conditioning; *observer-relativity of facts* and the *movable cut* are theorems of an enlarged observer-context category; and the Frauchiger–Renner argument localizes to an ordinary state-dependent violation (contextual fraction exactly 1/6) carried by a single CHSH facet, with the loop holonomy of the observer switch provably trivial. Two of these signatures have been measured on superconducting hardware. Each tenet is either a theorem of the companion papers, a machine-verified certificate in the public suite, or is explicitly labeled a conjecture. Version v1 of this note contained four errors of substance; §2 states them and their repairs.

1 Introduction: the principle, restated

Textbook presentations of the Copenhagen interpretation bundle together several commitments: that classical concepts apply exactly within a single experimental arrangement; that arrangements exclude one another (complementarity); that measurement updates the state by projection; that facts are relative to the experimental context, including the observer; and that the boundary between quantum system and classical apparatus — the cut — can be moved without empirical consequence. Critics have long held that these are attitudes rather than assertions. This note argues the opposite: each commitment is a precise structural statement about a single mathematical object, and each is now either proved, machine-certified in a public verification suite, or stated as an explicit conjecture with its current evidence.

The object is the transformation groupoid

$$\mathcal{G} \ = \ \mathrm{PU}(n) \ltimes X,$$

where X is the space of maximal abelian $*$-subalgebras (MASAs, i.e. measurement contexts) of $M_n(\mathbb{C})$ and the projective unitary group acts by conjugation, together with the canonical $\mathrm{U}(1)$ 2-cocycle obtained from the central extension $\mathrm{U}(1) \to \mathrm{U}(n) \to \mathrm{PU}(n)$. The formulation is then:

Principle 1 (Copenhagen, structural form)**.** *(i) Local validity.* Every single context is exactly classical: the restriction of the canonical cocycle to any one context trivializes, with an explicit trivializing cochain. *(ii) Complementarity.* The cocycle does not trivialize globally; its holonomy along closed context loops is a quantized obstruction, and this obstruction is the all-versus-nothing sector of contextuality, with the state sector governed by an enumerable facet geometry. *(iii) Collapse.* Statistics, repeatability, and covariance under the measured observable's commutant pin the measurement update to a one-parameter family per degenerate block, with the Lüders rule its unique efficient member; classically the same axioms force Bayesian conditioning up to the same per-cell parameter. *(iv) Relative facts.* Value assignments exist only per context; the obstruction persists under enlargement of the observer. *(v) The movable cut.* The placement of the system/apparatus boundary is a choice of chart; section-fixed comparisons of differently-cut descriptions agree.

Throughout, "verified" means: reproduced by the named script in the companion repository [5], with the expected one-line output pinned in `verification/INDEX.md`. The supporting mathematics is developed in the companion papers [1, 2, 3, 4]; this note is the synthesis; proofs live in the companion papers and the verification supplement, with the source of every numbered claim listed in Appendix A.

2 What changed since v1

Version v1 (February 2026) contained four errors of substance, found by adversarial review and machine checking. We state them because the credibility of a structural program rests on its error ledger.

1. *Wrong groupoid.* v1 built the cocycle on $\mathrm{U}(n) \ltimes X$. The lifting class there is vacuous (Theorem 1 of [1]; `thmG_general.py`); the canonical class lives on $\mathrm{PU}(n) \ltimes X$, where it has order exactly n [1, Thm. 8].

2. *Overclaim "encodes".* v1 asserted that holonomy *encodes* contextuality. False in both directions: at $d = 3$ the class is nontrivial of order 3 while the achievable state-independent all-versus-nothing value is 0 (`d3_gap_certificate.py`), and KCBS contextuality is invisible to the class (`kcbs_converse.py`). The correct statement is *detects the AvN sector* (Theorem 3 below).

3. *Level-unspecified Wigner–friend holonomy.* v1 attributed $-i$ to "the" holonomy of the friend loop. The projective (ray-level) holonomy is $+1$; $-i$ is the determinant-*section* holonomy, with section-independent square -1 (`wf_loop_holonomy.py`). Claims must pin the level.

4. *Efficiency-restricted Lüders uniqueness.* v1's uniqueness argument for the update rule quantified only over efficient (single-Kraus) instruments. The instrument-level theorem without that restriction is Theorem 7 below.

Corrections (2026-07-11, second external audit). A follow-up adversarial pass found further defects in the measurement-update layer, now repaired (the core obstruction-spectrum results of [1, 2] are unaffected): (a) the collapse family's complete-positivity range was misstated — on a rank-r block it is the affine interval $-1/(r^2-1) \leq p \leq 1$, not $[0, 1]$ (Theorem 7); (b) the sharp-core counterexample $\mathrm{diag}(1, c)$ was invalid (its second outcome is not repeatable) and is replaced by a qutrit example (Theorem 8); (c) the classical twin's range is $-1/(m-1) \leq t \leq 1$ with a second deterministic member (the swap) at $m = 2$ (Theorem 9); (d) the V37 evidence is a numerical stress test, not a "machine-exact" certificate; (e) the observer structure of [4] is a *category* with a switch subgroupoid, not a groupoid; and (f) the hardware significance is statistical (shot-noise) under the implemented witness model, conditional on compatibility/nondisturbance assumptions.

3 Local validity is exact

Theorem 1 (Local classicality). *Let C be any single Weyl context at local dimension d. The restriction of the canonical cocycle to C (the Weyl cocycle on the context's label group) is a coboundary, with a trivializing cochain λ valued in μ_d for* even d *and in μ_{2d} for* odd d*; the odd-d bound is sharp (an explicit $d = 3$ group requires μ_{2d}).*

Proof sketch. On a commuting label group G the parity of the τ-exponent $c(u, v)$ equals $\langle u, v\rangle$ mod 2, which vanishes for even d; hence $c = 2c'$ with c' a symmetric $\mathbb{Z}_d$-valued 2-cocycle. Symmetric classes form $\mathrm{Ext}(G, \mathbb{Z}_d)$, additive over a cyclic decomposition of G and detected by the cyclic transgressions $\sum_{j<a} c'(jg, g) = \frac{1}{2}a(a-2)q(g) - dK$, which are divisible by $a = \mathrm{ord}(g)$ in both parity cases; every cyclic class therefore vanishes, and so does the total class. For odd d the same transgression argument at level $2d$ gives μ_{2d}. Certified, with complete searches and the $d = 3$ sharpness witness, in `mu_d_valuation.py` (V46). □

An immediate corollary of the parity lemma: at even d every commuting closed context has *even* τ-level phase — context products are always ω-powers, and integrality is automatic rather than a gauge choice. (An earlier claim that half-steps $\tau \notin \mu_d$ are necessary at even d traced to a label-layout artifact in a verification harness and is retracted; see V44/V46 in the suite index. Genuine τ-steps are an odd-d phenomenon. The *state-sector* τ-necessity results of [3] are a different statement and are unaffected.) Explicit per-context trivializers: `local_classicality.py` (V23), for all rows and columns of the Peres–Mermin square at $d = 2$, a $d = 3$ sample family, and the certified $d = 4$ family; the same script exhibits the global contrast (no joint λ exists across contexts where the obstruction is present, cross-referencing V10/V13/V33). This is tenet (i) of Principle 1 in exact form: every chart of the atlas is flat, with the flattening written down.

4 The global twist: what holonomy detects

Theorem 2 (Canonical class; [1]). *On $\mathrm{U}(n) \ltimes X$ the $\mathrm{U}(1)$ lifting class vanishes identically. After Morita reduction, the canonical class on $\mathrm{PU}(n) \ltimes X$ has order exactly n.*

Theorem 3 (Detection equivalence). *Let F be a closed-triple Weyl context family at even d, with incidence matrix A and context-phase vector $s \in \mathbb{Z}_d^{\mathrm{ctx}}$ (canonical: by the parity lemma of Theorem 1, commuting context phases at even d are automatically ω-powers). The following are equivalent: (a) no noncontextual value assignment over $\mathbb{Z}_d$ exists; (b) some left-kernel vector u*

($uA \equiv 0 \bmod d$) has $u \cdot s \equiv d/2 \pmod d$; (c) some certificate carries odd commutator carry [2, Thm. 4]. The achievable state-independent values are exactly $\{0, d/2\}$ for even d and $\{0\}$ for odd d [2, Thm. 1].

Proof. (b)⇒(a): a solution γ would give $u \cdot s \equiv (uA)\gamma \equiv 0$. (a)⇒(b): by Smith normal form over $\mathbb{Z}$, unsolvability of $A\gamma \equiv s \pmod d$ produces, constructively, $u = \big(d/\gcd(g_i, d)\big)U_i$ with $uA \equiv 0$ and $u \cdot s \not\equiv 0 \pmod d$ (double annihilation over $\mathbb{Z}_d$); such a u is a certificate of the multiplicity covered by [2, Rem. 3], so by the Obstruction Spectrum Theorem [2, Thm. 1] its value satisfies $2(u \cdot s) \equiv 0$, forcing $u \cdot s \equiv d/2$. (b)⇔(c) is [2, Thm. 4]. Finally, no gauge hypothesis is needed: by the parity lemma of Theorem 1, every commuting closed context at even d has even τ-level phase, so $s = s^{(2d)}/2 \in \mathbb{Z}_d^{\mathrm{ctx}}$ is canonical. (For general closed-triple data — commuting or not — the q-parity identity $s^{(2d)} \equiv A\bar{q} \pmod 2$, $\bar{q}_v = q(v) \bmod 2$, obtained by telescoping the Weyl product law, supplies the constructive gauge; kernel pairings are gauge-invariant.) Certified end-to-end, with random and pinned positives, in `solvability_equivalence.py` (V44, layout-corrected), `tau_gauge_general.py` (V45, including the $d = 6$ extension), and `mu_d_valuation.py` (V46). □

What was, in v2, a machine-verified converse (2,100 families, zero mismatches) is thus a corollary of [2]'s proven theorems plus five lines of ring theory — the gap was assembly, not mathematics. Both containments of the naive slogan fail, and strictly (`d3_gap_certificate.py`, `kcbs_converse.py`): the class does not imply operational contextuality, and state-dependent contextuality can be class-invisible. Complementarity, tenet (ii), is therefore *located*: it is the AvN obstruction, no more and no less at the level of the canonical class.

4.1 The state sector: a facet geometry, now enumerated

State-dependent contextuality is quantified by the contextual fraction CF. The following results replace v1's vague appeal to "holonomy of the state" with exact geometry.

Theorem 4 (Codeword-polytope classification, $d = 2$; V31/V32)**.** *For closed-triple Weyl families at $d = 2$, $\mathrm{CF}(F, \rho) = 0$ if and only if the correlation vector $c(\rho)$ lies in the convex hull of the codeword set $L(F)$; quantitatively $\mathrm{CF} = \max\big(0, (S^* - 1)/2\big)$, where S^* is the optimal facet value, proved end-to-end via a completion lemma and Abramsky–Barbosa–Mansfield duality.*

Proposition 1 (On-hull)**.** *Every quantum behavior of the certified five-context $d = 4$ family of Theorem 5 satisfies three families of exact linear identities: per-context normalization, shared-observable marginal consistency, and vanishing on spectrally forbidden joint outcomes (24 constraints). The affine space they cut out has dimension exactly 33, equal to the affine span of the 64 deterministic behaviors, which satisfy the same identities. Hence quantum moment vectors lie in the deterministic affine hull.*

Theorem 5 (Complete facet census, certified $d = 4$ family; V43)**.** *For the certified five-context $d = 4$ family with one context deleted, the noncontextual polytope has exactly 23,256 facets in 61 orbit classes under the derived vertex-symmetry group of order 768. By Proposition 1, $\mathrm{CF}(\rho) > 0$ if and only if $\mu(\rho)$ violates one of the 23,256 inequalities.*

The census was derived twice by independent methods — an adjacency decomposition on the facet–ridge graph with every pivot certified in exact integer arithmetic, and a direct exact enumeration by NORMALIZ — which agree class-for-class (`d4_facet_census.py`). Structural laws of this geometry are proved in [3]: the ghost–tower law $\gamma - s^* \equiv d/2$ for context deletions, and the necessity of the τ-level (half-step) data for optimal separation (`d4_moment_facets.py`).

Conjecture 1 (Scenario-paired classification)**.** *For every closed-triple Weyl family at even d, $\mathrm{CF}(F,\rho) = 0$ iff $c(\rho) \in \operatorname{conv} L(F)$ in the scenario-paired formulation of [3].*

The global-frame formulations of this statement are refuted (V28–V30); the pairing is forced, not optional. The status is now sharply split by the choice of paired coordinates (V48/V52). In the *full-character* reading ($c(\rho) =$ all per-context characters), the classification is a *theorem*: the support lemma confines outputs to the spectrally allowed set, the paired characters determine each context's distribution by Fourier completeness, and an LP mass identity converts hull membership into an exact mixture — in this reading the statement is near-definitional, which is why the 5,556-record harness (V48; $d = 4, 8$, census-arbitrated, zero mismatches) could not fail: it certifies implementations. The substantive open form is the *compressed* reading (the correlator-type coordinates of [3], whose $d = 2$ case is Theorem 4's actual content). Its residual is *not* an on-hull rank identity: the natural generalization of Prop. 1 to arbitrary families is *false* (V52: pinned families with affine-dimension gaps up to 74 at $d = 8$; the census family's rank equality is special). The correct reduction (proven, V52): quantum behaviors lie in the deterministic affine hull iff a finite per-family *phase-coherence law* holds — characters with equal restriction to the section lattice carry a single Weyl label and phase; the label half is proven (Smith-form annihilator plus closure), and the phase half — a cross-context τ-holonomy congruence of the same species as the q-parity identity — is the named open lemma, exactly verified on all 36 tested families at $d = 4, 6, 8$.

Theorem 6 (Tower holonomy, resolved by cases; V47)**.** *(i)* Operator level: *every closed Weyl word has holonomy in μ_{2d}, and every value of μ_{2d} is attained (proved for even d; verified instances at odd d).* *(ii)* Determinant-section level *(canonical τ-grid section): at even d all loop holonomies lie in μ_{2d}; the attained group equals μ_{2d} exactly when $d \equiv 2 \pmod 4$, and is* constant μ_4 *along the 2-adic tower $d = 2, 4, 8, 16$.*

The original conjecture's exhaustiveness clause is thus *refuted*: the coincidence $\mu_4 = \mu_{2d}$ at the base $d = 2$ does not propagate up the tower. At odd d the μ_{2d} bound is section-dependent, and both directions are now *proven* (V51): the DFT section violates it exactly when $d \equiv 3 \pmod 4$ (Schur's determinant $\det F = i^{d(d-1)/2}$; pinned instance: the $d = 3$ observer loop has holonomy $e^{i\pi/6}$), while in the Weil-normalized section every loop holonomy lies in μ_{2d} for all odd d — the Gauss phase cancels the μ_4 part of Schur's phase exactly ($\det F_W = (-1)^{\lfloor d/4 \rfloor}$), $\det M_a$ is the Jacobi symbol $(a|d)$ by Zolotarev, and the Weil-attained group is μ_6 if $3 \mid d$ and μ_2 otherwise, exhaustive only at $d = 3$. Claims must pin the section — the same lesson as v1's error (3).

5 Collapse: the admissible update rule, and its classical twin

Theorem 7 (Instrument-level Lüders; V37)**.** *Let $\{\mathcal{I}_j\}$ be an instrument for a projective measurement with degenerate blocks, satisfying (i) statistics, (ii) repeatability, and (iii) covariance under the commutant of the measured observable. Then on each degenerate block of rank r_j,*

$$\mathcal{I}_j(\rho) = p_j\, P_j \rho P_j + (1 - p_j)\, \mathrm{tr}(P_j \rho)\, \frac{P_j}{r_j}, \qquad -\frac{1}{r_j^2 - 1} \le p_j \le 1,$$

an affine *depolarizing family (the lower endpoint is forced by complete positivity; this is* not *only the convex segment $p_j \in [0,1]$). The linear solution space has exactly one parameter per degenerate block (nullity* 2 *at* $(4,[2,2])$*, verified also at* $(6,[3,2,1])$*), and the Lüders instrument $p_j = 1$ is its unique efficient (single-Kraus) member. Covariance under the MASA alone is strictly insufficient. Proof via Schur's lemma on block-covariant channels together with the Choi bound; the V37 script is a numerical affine-dimension stress test, and the CP interval is certified exactly in* `lueders_cp_interval.py`.

Theorem 8 (Sharp core; V40)**.** *For a POVM instrument, statistics plus repeatability force every output of $\mathcal{I}_j$ into the eigenvalue-*1 *subspace of its effect E_j, and these subspaces are mutually orthogonal. A strictly unsharp effect (no* 1*-eigenvector) admits no repeatable instrument at all. The folk claim "repeatable ⇒ projective" is nevertheless false as stated, but a valid witness must be at least three-dimensional: the qutrit two-effect POVM* $E_1 = \mathrm{diag}(1,0,c)$*,* $E_2 = \mathrm{diag}(0,1,1-c)$*,* $0 < c < 1$*, admits a repeatable covariant instrument on* every *outcome while E_1 is non-projective. (An earlier qubit example* $\mathrm{diag}(1,c)$ *was withdrawn: its second effect* $\mathrm{diag}(0,1-c)$ *has no eigenvalue* 1*, so that outcome is not repeatable.)*

Theorem 9 (Classical twin; V40)**.** *The same axioms, interpreted for classical stochastic instruments on a finite outcome partition, force the affine family $T_t = t\,I + (1-t)\,J/m$ per cell of size m, with $-\frac{1}{m-1} \leq t \leq 1$ (nonnegativity), the nullities matching the quantum case block-for-block. Its deterministic covariant members are the identity ($t = 1$, Bayesian conditioning) for every m and, in addition, the swap ($t = -1$) when $m = 2$; so Bayes is the unique deterministic member only for $m \geq 3$.*

Remark 1 (Position)*.* Theorems 7–9 give the Leifer–Spekkens thesis a sharpened form: *conditional on* statistics, repeatability and full commutant covariance, the admissible update rule is restricted to an affine depolarizing family whose unique single-Kraus member is Lüders, with Bayes' rule as the classical shadow of the same axioms. This is a characterization of admissible instruments — not a derivation that collapse occurs, nor a resolution of the definite-outcome problem; rival interpretations can use the same formal result. What is quantum is not the updating; it is the geometry (§4) on which the updating lives.

6 Observers, relative facts, and the cut

The observer enters through the observer-context category of [4], whose objects are pairs (context, observer chain) and whose morphisms are switches within a fixed enlargement (isomorphisms, forming a subgroupoid) together with refinement isometries into larger Hilbert spaces (non-invertible); the presence of non-invertible refinements makes this a *category*, not a groupoid, and holonomy invariants are taken on the switch subgroupoid.

Theorem 10 (Relative facts; [4])**.** *Value assignments exist per object and pull back monotonically along fibered morphisms; the global obstruction persists under Wigner enlargement — for the Peres–Mermin square tensored with a friend ($d = 8$), the context signs remain* $(+1)^5(-1)^1$ *(*`observer_groupoid.py`*, certificate A).*

Theorem 11 ($\mathbb{Z}_2$ switching invariant; [4])**.** *The square of the switch-loop holonomy is a gauge-independent $\mathbb{Z}_2$ invariant (proved via Weyl determinants* ±1*):* -1 *on the mutually-unbiased-bases triangle,* $+1$ *on within-context Weyl loops. The converse fails: a complementary* 2*-cycle with value* $+1$ *is pinned.*

Theorem 12 (Movable cut; [4])**.** *Section-fixed comparisons of descriptions differing by cut placement agree, with the update rule of Theorem 7 forced on both sides.*

Theorem 13 (Frauchiger–Renner localized; [4])**.** *In the observer-context category, the FR cycle carries trivial class: all switch-loop phases are* $+1$*, fully gauge-invariantly. The paradox localizes to an ordinary state-dependent violation:* $\mathrm{CF} = 1/6$ *exactly, attained on the single CHSH facet* $\langle \bar{Z}Z \rangle - \langle \bar{Z}X \rangle + \langle \bar{X}X \rangle + \langle \bar{X}Z \rangle \leq 2$ *at quantum value* $7/3$ *(`fr_cycle.py`).*

Theorem 13 *refutes* v1's implicit suggestion (and [4]'s own initial conjecture) that the FR argument is a holonomy phenomenon: the switch holonomy is trivial; what FR exercises is the state sector of §4.1, with definite coordinates and a definite price.

The Wigner–friend loop, level-pinned. For the loop $T \to T_x \to T_y \to T$ generated by $\mathbb{1} \otimes H, \mathbb{1} \otimes S$: the ray-level (Pancharatnam) holonomy is $+1$; the determinant-section holonomy is exactly $-i$, a τ-level invariant with section-independent square -1 (`wf_loop_holonomy.py`).

Hardware. On `ibm_fez` (job `d986q62f47jc73a7hm2g`): the interferometric loop phase is $-2.93° \pm 1.55°$, consistent with ray-level $+1$ and excluding $-i$ at $\sim 56\sigma$; on the same device the Peres–Mermin witness attains $S = 4.6125 \pm 0.0173$, i.e. 35.4σ above the tight deterministic bound 4 (`hardware/results/ibm_fez_holonomy_20260709.json`). One machine, both signatures, both sharp.

7 The principle, assembled

Principle 1 now reads as a theorem schedule. (i) is Theorem 1. (ii) is Theorems 2–3 for the state-independent sector and Theorems 4–5 for the state sector, with Conjecture 1 (now with $d = 4, 8$ evidence, V48) and the case-resolved tower Theorem 6 marking the frontier. (iii) is Theorems 7–9. (iv) is Theorems 10–11 and the localization Theorem 13. (v) is Theorem 12. Nothing here modifies quantum predictions; the claim is that the Copenhagen commitments, correctly stated, are structure rather than stance — local flatness, one quantized global twist, forced bookkeeping, chart-relative facts, and chart-choice for the cut.

Open Problem 1. *(a) The cohomological home of class-invisible (KCBS-type) contextuality. Progress (V49/V50): support-presheaf and scenario-complex cohomologies provably cannot capture it (pinned no-gos); a* $\mathrm{U}(1)$*-holonomy-twisted spectral pairing on the non-orthogonality graph characterizes the state sector exactly on odd cycles (*C_5, C_7, C_9*:* $\mathrm{CF} = 2\max(0, S - \alpha)$*, all holonomy coordinates active, trivial class Perron-maximal) and on the* $\alpha = 2$ *antiholes; the once-mysterious "*(-1)*-twist = classical bound" is the reflection identity* $G \mapsto 2I - G$ *and holds exactly when* $\alpha = 2$*. The Peres–Mermin milestone is achieved (V54): on the* 24*-ray configuration the same local system is Kochen–Specker–rigid with pure-torsion holonomy, and the AvN* -1 *is its* forced *holonomy invariant (transversal parity theorem;* $h(z_{\mathrm{obs}}) = -1$*), with flex dimension —* 0 *for PM,* 2 *for KCBS — as the switch between the two layers. The general dictionary (flexes = state moduli; rigidity + forced torsion ⇔ AvN) remains conjectural. (b) The general-d facet census. The* $d = 4$ *arithmetic profile is complete (V53):* 18 *species, all facet functionals rational (the tier-2 "*$4\sqrt{2}$*" unit bounds were a normalization artifact — exact* $\{0, \pm 1\}$ *lifts with integer bounds* $\{6, 7\}$ *exist),* τ*-level data necessary for all* 61 *classes; the general-d census itself needs Normaliz-class compute (protocol pinned in V53). (c) Conjecture 1 in the compressed* *reading: the Θ-coherence congruence of V52 (its label half is proven; the odd-d Weil-section case of Theorem 6 is now closed, V51).*

A Claims-to-verification table

Claim	Source	Verification
Thm. 1	this note	`local_classicality.py` (V23); `mu_d_valuation.py` (V46)
Thm. 2	[1] Thm. 1, 8	`thmG_general.py`
Thm. 3	this note; [2] Thm. 1, 4	`solvability_equivalence.py` (V44); `tau_gauge_general.py` (V45)
Thm. 6	this note	`tower_holonomy.py` (V47); `weil_section_theorem.py` (V51)
Conj. 1 (split status)	this note; [3]	`scenario_paired_evidence.py` (V48); `paired_classification_theor`
gaps (both strict)	[3]	`d3_gap_certificate.py`, `kcbs_converse.py`
Thm. 4	[3] Thm. 2	`paired_frame_construction.py`, `ghost_facet_theorem.py`
Thm. 5	this note; [3]	`d4_facet_census.py` (V43)
tower laws	[3]	`d4_ghost_tower.py`, `d4_moment_facets.py`
Thm. 7	this note	`lueders_instrument.py` (V37)
Thm. 8, 9	this note	`s4_povm_bayes.py` (V40)
Thm. 10, 11	[4]	`observer_groupoid.py` (V38)
Thm. 12	[4]	V37 + [4] §3
Thm. 13	[4]	`fr_cycle.py` (V39)
WF level split	[3]	`wf_loop_holonomy.py`
Prop. 1	this note	`d4_facet_census.py` (exact-rank lemma)
S4/S5 proofs	v2 supplement	`note_v2_patch.md`
hardware	—	`ibm_fez` job `d986q62f47jc73a7hm2g`

References

[1] M. Flores Gordillo, *The Canonical Class of the MASA Context Stack and the Peres–Mermin Obstruction*, working draft, 2026.

[2] M. Flores Gordillo, *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower*, working draft, 2026.

[3] M. Flores Gordillo, *What Contextual Holonomy Detects*, working draft, 2026.

[4] M. Flores Gordillo, *The Observer-Context Category: Relative Facts, a $\mathbb{Z}_2$ Switching Invariant, and the Movable Cut*, working draft v0.3, 2026.

[5] `github.com/manuflog/contextuality-obstructions`, verification suite, 2026.